Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Abstract

Glossary of acronyms

Acronym	**Meaning**
CCGT	Combined-cycle gas turbine: a flexible, dispatchable thermal generator.
DA	Day-ahead wholesale electricity market, cleared once per day.
IDA	Intraday auctions: sequence of (European) auctions held after the day-ahead clearing.
MTU	Market time unit: size of the time interval at which bids are submitted and prices clear.
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the market operator for the Spanish and Portuguese wholesale electricity markets.

1 Introduction

1.1 Motivation and research question

1.2 Related literature

1.3 Mechanism summary and contribution

TWO DISTINCT PURPOSES OF MTU15:

- OPERATIONAL / SYSTEM-OPERATOR CHANNEL. LETTING THE SYSTEM FOLLOW WITHIN-HOUR SOLAR VARIABILITY (CLOUD COVER, RAMP). MIDDAY IS EXACTLY WHERE THIS MATTERS MOST.
- STRATEGIC-EXPLOITATION CHANNEL. LETTING PIVOTAL FIRMS RE-PRICE ACROSS QUARTERS WHEN NET-LOAD MOVES SYSTEMATICALLY WITHIN THE HOUR. THIS IS WHAT DOMINANT CCGT CAN MONETIZE. **THIS IS WHAT WE DO!!**

1.4 Roadmap

2 Theoretical model

2.1 Setup

2.2 Equilibrium under low granularity

2.3 Equilibrium under high granularity

2.4 Predictions: dominant vs fringe response in time-varying vs flat subperiods

TODO: COROLLARY ON RAMP-CONSTRAINED HOURS. WHEN THE SYSTEM OPERATOR IMPOSES BINDING RAMP RESTRICTIONS (QUANTITY CANNOT ADJUST MUCH ACROSS CONSECUTIVE QUARTERS), THE HIGH-GRANULARITY REGIME FORCES FIRMS TO AMPLIFY THEIR STRATEGIC RESPONSE ON THE PRICE MARGIN RATHER THAN THE QUANTITY MARGIN. THE MTU15 REFORM THEREFORE AMPLIFIES MARKET POWER IN PRECISELY THOSE HOURS WHERE PHYSICAL ADJUSTMENT IS CONSTRAINED.

3 Institutional setting, data, and identification

3.1 Spanish wholesale electricity market: a sequential-market overview

STYLISED MARKET DIAGRAM TBD HERE AS TIKZ: DA - IDA AUCTIONS - CONTINUOUS - BALANCING

3.2 The MTU15 reform sequence

3.3 Data sources

OMIE

- Day-ahead data:

- Intraday-auction data:
- Continuous intraday market data:

ESIOS
ENTSO-E
INCLUDE SAMPLE SIZE AND FAMILIES.

3.4 Critical-vs-flat hours: empirical calibration

Our identification strategy relies on the idea that the reform does not give extra strategic space for firms to exploit in all hours (see Section 2.4). The reform's strategic effect is concentrated in hours where the *residual demand* (electricity demand minus wind and solar production) varies substantially within the hour — those are the hours where finer time resolution allows firms to set quarter-specific bids that match the within-hour conditions.

Classification criterion. For each clock-hour h, define the **within-hour variance of residual demand**,

$$\sigma^2_{\text{within}}(h) \;=\; \mathbb{E}_d\left[\tfrac{1}{4}\sum_{q=1}^{4}\big(D_{h,q,d}-\bar{D}_{h,d}\big)^2\right], \qquad D_{h,q,d}=\text{demand}_{h,q,d}-\text{wind}_{h,q,d}-\text{solar}_{h,q,d}, \quad (1)$$

where $\mathbb{E}_d$ denotes the empirical expectation taken across days d and the variance is taken across the four 15-minute quarters q for a particular clock-hour h. **Critical hours** are those with high σ^2_{within} (large within-hour movement in residual demand, in either direction); **flat hours** are those with low σ^2_{within}. Borderline hours (moderate variance, no clean class assignment) are dropped from the analysis.

Applying this criterion to Spain 2025 yields the following grouping:

Pattern	**Hours (local)**	**Mechanism**	**Class**
Overnight valley	01:00–04:00	Demand low and stable; no solar; system slack	*flat*
Morning ramp + solar entry	05:00–09:00	Demand rises sharply (1.5–1.8 GW within the hour); solar starts entering between 06:00 and 09:00	*critical*
Midday solar plateau	11:00–15:00	Demand flat at its daytime level; solar near peak but variable across days (cloud cover)	dropped
Afternoon ramp + solar exit	16:00–20:00	Demand rises into the evening peak; solar production fades	*critical*
Post-peak descent	20:00–23:00	Demand falls sharply from the evening peak (1.5–1.9 GW within the hour); solar already gone	*critical*
Transitional	00:00, 04:00, 09:00, 10:00, 15:00, 23:00	Mixed/small variation; no clean class assignment	dropped

The critical-hour set used in the empirical analysis is 05:00–09:00 and 16:00–23:00 (eleven clock-hours). The flat-hour set is 01:00–04:00 (three clock-hours). The full 24-hour ranking by σ^2_{within}, mean demand, mean solar, and within-hour demand change is reported in Appendix B.6.

OJO, IF WE INTRODUCE THE 11:00-15:00 PERIOD IN THE CONTROL, WE ARE MIXING THE OPERATIONAL AND STRATEGIC CHANNELS OF THE REFORM, BECAUSE THE OPERATIONAL EFFECT IS CONCENTRATED IN THOSE HOURS!!!

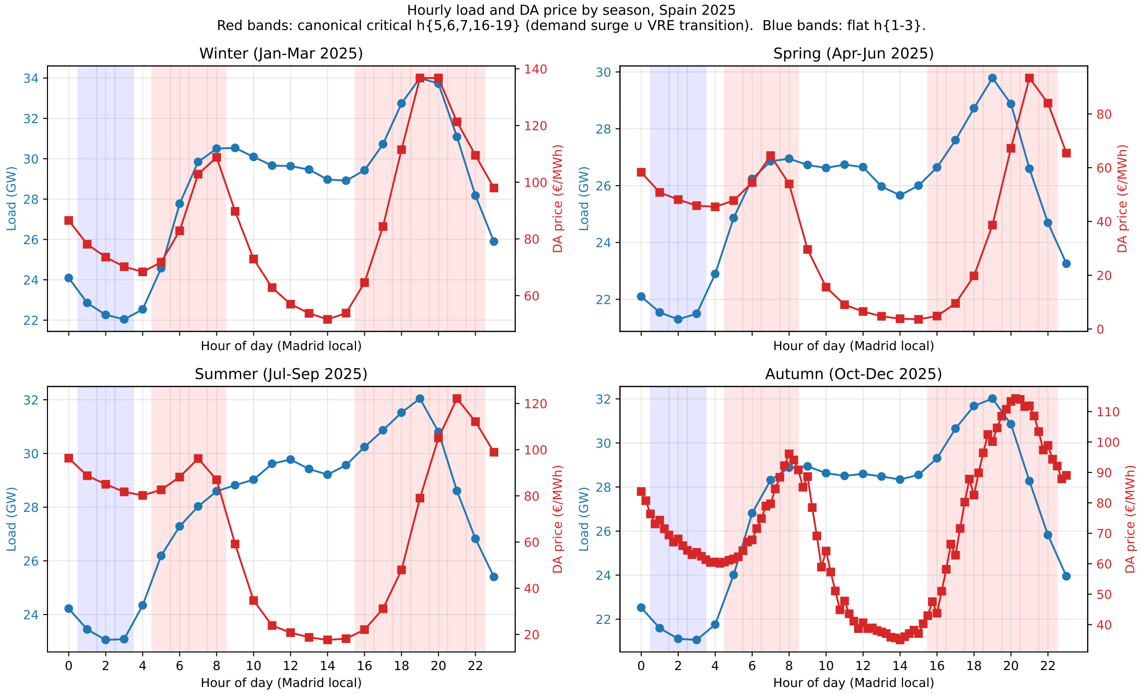


Figure 1: Hourly load and DA price profile, by season, Spain 2025. Four-panel decomposition by season. Blue line: realized Spanish electricity demand; the slope of the line is the within-hour demand ramp rate. Red line: day-ahead clearing price. Red bands mark the *critical hours* (05:00–09:00 and 16:00–23:00, local time): high within-hour variance of residual demand. Blue bands mark the *flat hours* (01:00–04:00): low and stable demand, no solar, system slack. Critical hours combine the morning ramp, the afternoon transition into the evening peak, and the post-peak descent; flat hours are the overnight valley.

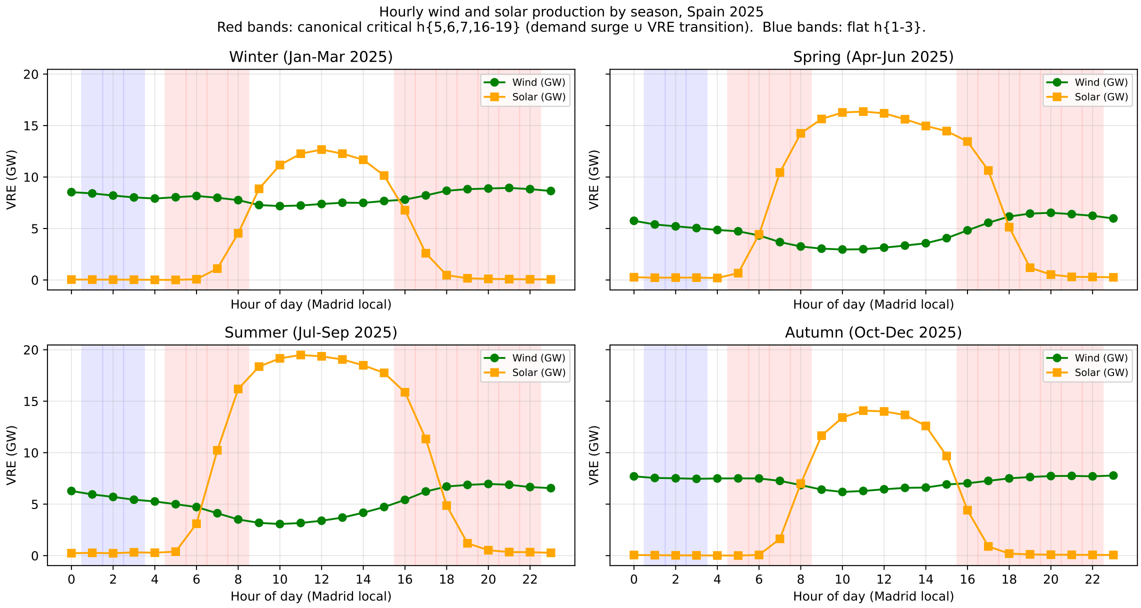


Figure 2: Hourly wind and solar production by season, Spain 2025. Four-panel decomposition by season. Red bands mark the *critical hours* (05:00–09:00 and 16:00–23:00). Blue bands mark the *flat hours* (01:00–04:00). Solar production is concentrated in midday hours and fades sharply at the boundaries of the critical-hour bands; wind is broadly flat across the day with mild seasonality. The combination implies that critical hours systematically coincide with low renewable supply year-round, reinforcing the case for treating them as the strategically relevant time-varying window.

3.5 Firms in the analysis sample

MENTION THAT WE USE LISTA_UNIDADES AND LISTA_AGENTES FROM OMIE, AS WELL AS THE SPECIFIC FAMILIES FORM OMIE, WHICH HAS FIRM AND UNIT INDICATORS. ALSO IMPORTANT TO MENTION THE CNMC CLASSIFICATION OF DOMINANT (SOURCE)

The empirical analysis follows nine parent firms in two blocks. The first block contains the *dominant operators in Spanish generation* (the CNMC *operador-dominante-en-generación* tier): Iberdrola, Endesa, Naturgy and EDP-Spain. EDP-Portugal joins this block as the dominant Iberian generator on the Portuguese side of MIBEL. The second block contains four other large vertically-integrated operators – Repsol, Engie España, TotalEnergies, and Moeve – which retain substantial portfolios but are non-dominant in the CNMC sense. Whether the firms in each block actually have a strategic margin in critical hours is an empirical question that we answer in Section 4.1, examining pivotality on the day-ahead bid data.

Table 1: Firms in the analysis sample. Capacity is the sum across units of the maximum power offered in any 2025 day-ahead period (a tight lower bound on installed nameplate), share-weighted across joint-owned plants. Market share is the firm's share of total Iberian day-ahead programmed sell volume in 2025 from PDBF, combining the DA auction and bilateral contract channels and share-weighted across joint-owned plants. The breakdown by channel (auction vs bilateral vs IDA), unit counts, and spot-market net position is reported in Appendix B.2.

Parent firm	Main generation technologies	Capacity (GW)	Market share (%)
Dominant operators in Spanish generation (CNMC tier)			
Iberdrola	Solar, Hydro, CCGT	31.03	24.94
Endesa	Hydro, CCGT, Nuclear	12.46	16.39
Naturgy	CCGT, Hydro, Solar	12.60	12.34
EDP-Spain	Wind, Hydro, Solar	4.73	4.14
EDP-Portugal	Hydro, CCGT	7.42	15.17
Other large operators (non-dominant)			
Repsol	Solar, Hydro	0.61	1.43
Engie España	Wind, Solar, CCGT	4.05	1.73
TotalEnergies	Solar, CCGT, Wind	1.19	0.63
Moeve	Solar, Hydro, CCGT	1.28	0.01

3.6 Identification strategy and treatment-effect set

WE SHOULD DEFINE HERE HOW WE DEFINE PIVOTALITY AND HOW WE CALCULATE IT, WITH DISCUSSION.

ALSO WE SHOULD DISCUSS THAT THE TREATMENT IN OUR IDENTIFICATION IS AT THE HOUR LEVEL (CRITICAL VS FLAT), NOT AT THE FIRM LEVEL, AND THAT FIRMS ENTER THE ANALYSIS AS SUBJECTS: PIVOTAL FIRMS ARE THOSE EXPECTED TO REACT TO THE TREATMENT BECAUSE THEY HAVE A STRATEGIC MARGIN IN CRITICAL HOURS; NON-PIVOTAL FIRMS SERVE AS PLACEBO BECAUSE THEY BID AT MARGINAL COST REGARDLESS OF HOUR.

HOWEVER, WE SHOULD THINK IF THIS IS THE BEST WAY TO PRESENT IT, MAYBE IT IS BETTER TO TALK ABOUT "EXPOSURE" TO THE TREATMENT ACROSS TIME OR SOMETHING LIKE THAT!! TO THINK ABOUT IT

3.7 Parallel-trends discussion

IMPORTANT THING HERE IS TO DISCUSS IT FOR EACH OF THE OUTCOMES AND TO TALK ABOUT CONDITIONAL PARALLEL TREND WITH RENEWABLE CAPACITY CONTROLS!

4 Bidding-behavior evidence of the mechanism

4.1 Pivotality: which firms can exploit the granularity reform

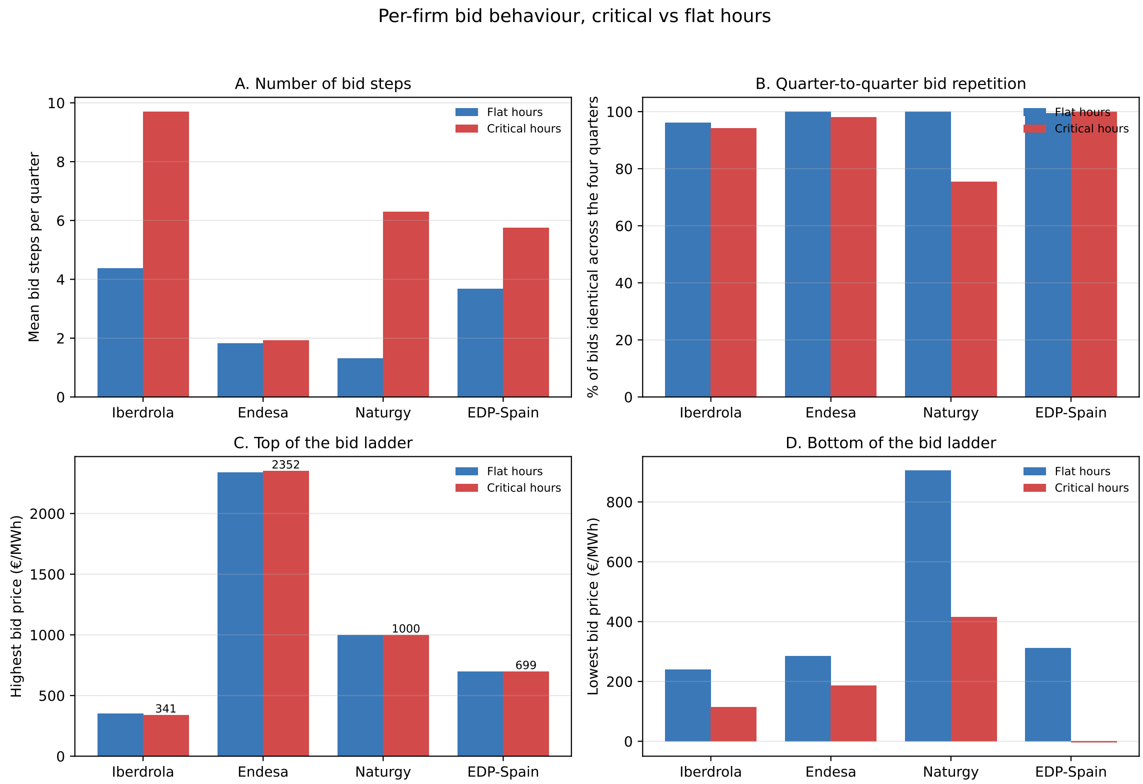
OJO, HYDRO DOES NOT EXPLOIT GRANULARITY BECAUSE THE DISPATCH DECISION IS DOMINATED BY RESERVOIR WATER-VALUE OPTIMIZATION (A MULTI-DAY PROBLEM), NOT WITHIN-HOUR STRATEGY. SO THE GRANULARITY REFORM'S "STRATEGIC EXPLOITATION CHANNEL" HAS NO PURCHASE HERE EVEN THOUGH THE CAPACITY FOR MARKET POWER EXISTS!!!!

ONLY CCGT EXPLOITS GRANULARITY!! AND IN PARTICULAR NATURGY!

Table 2: Pivotality rate by parent firm and hour-class, CCGT, October–December 2025. Pivotality is the share of (date, unit, period) cells where the firm has a bid tranche within 1 EUR/MWh of the day-ahead clearing price (a strict price-setter test). A wider definition (any tranche below clearing combined with any tranche above) produces apparently large pivotality rates for any firm submitting a wide bid ladder, but those numbers do not reflect price-setting power. The strict measure used here corresponds to the firm being one marginal tranche away from setting the clearing price. Firms in the top panel are tightly pivotal in critical hours at meaningful rates relative to a market of ∼2,500 bidding units; firms in the bottom panel are essentially never tightly pivotal and serve as a negative control.

Firm (parent)	flat (01:00–04:00)	critical (05:00–09:00, 16:00–23:00)	Δ crit−flat
Tightly pivotal in critical hours (within 1 EUR/MWh of clearing)			
Naturgy	0.006	0.044	**0.038**
EDP-Spain	0.026	0.041	**0.015**
EDP-Portugal	0.005	0.028	0.023
Iberdrola	0.003	0.012	0.009
Endesa	0.001	0.004	0.003
Essentially never tightly pivotal (negative control)			
Engie España	0.000	0.000	0.000
Moeve (formerly Cepsa)	0.000	0.000	0.000
Repsol	0.000	0.000	0.000
TotalEnergies	0.000	0.000	0.000

4.2 Per-firm bid-shape and granularity exploitation



A bid step is one price-quantity pair in a firm's supply offer for a given period; a firm with more steps offers a finer-grained supply curve. 'Identical across quarters' = the firm submitted the exact same step (price and quantity) in all four quarters of the same clock-hour. Sample: combined-cycle gas (CCGT) units operated by the four largest Iberian groups, day-ahead market, October--December 2025 (post-reform window).

Figure 3: Per-firm bid behaviour, critical vs flat hours. Day-ahead bids submitted by the four largest CCGT operators in Spain (Iberdrola, Endesa, Naturgy, EDP-Spain), October–December 2025. Panel A: average number of price-quantity steps a firm includes in its bid for a 15-minute period. Panel B: share of bids that are identical (same price and same quantity) across all four quarters of the same clock-hour. Panel C and D: average highest and lowest priced step in the bid. Two distinct strategic patterns appear: Iberdrola enriches the ladder (more steps in critical hours) while keeping the four quarters of each hour nearly identical; Naturgy keeps the ladder shallow but varies its bids strongly from quarter to quarter (the mechanical-repeat rate drops by roughly 35 percentage points in critical hours).

5 Main empirical results

5.1 Strategic withholding and intraday upward repositioning

MECHANISM: UNDER THE HIGHER-GRANULARITY REGIME, A PIVOTAL DOMINANT FIRM IN A CRITICAL HOUR CAN WITHHOLD CAPACITY FROM THE DAY-AHEAD CLEARING AND REPOSITION THAT CAPACITY UPWARD IN THE INTRADAY AUCTIONS AT HIGHER PER-QUARTER PRICES. UPWARD SELL ADJUSTMENT SUMMED ACROSS THE THREE INTRADAY AUCTIONS, $q_{2,fdh}$, IN MWH PER UNIT-CLOCK-HOUR:

$$q_{2,fdh} \;=\; \alpha + \beta_1\,\mathrm{crit}_h + \beta_2\,\mathrm{post}_d + \beta_3\,(\mathrm{crit}_h \times \mathrm{post}_d) + \gamma_f + \delta_{\mathrm{DOW}(d)} + \varepsilon_{fdh}, \tag{2}$$

with firm fixed effects γ_f, day-of-week fixed effects $\delta_{\mathrm{DOW}(d)}$, and cluster-robust standard errors by date. The sample restricts to the same-calendar-month windows October–December 2024 (pre) and October–December 2025 (post), so the March 2025 intraday reform is held constant. Critical hours are 05:00–09:00 and 16:00–23:00; flat hours are 01:00–04:00. Pivotal firms are those with non-trivial price-setting capacity in critical hours (Iberdrola, Endesa, Naturgy, EDP-Spain, EDP-Portugal); non-pivotal firms (Repsol, Engie España, TotalEnergies, Moeve) serve as a negative control.

Table 3: B1 – Within-day DiD on q_2 (intraday upward adjustment). Estimates of Equation 2. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	All firms (1)	Pivotal firms (2)	Non-pivotal firms (3)
Critical (β_1)	1.280	-4.454	4.993
	(0.328)	(0.893)	(0.320)
Post (β_2)	-2.562	-8.294	1.140
	(0.486)	(1.146)	(0.415)
Critical × Post (β_3)	-1.397	3.866	-4.741
	(0.446)	(1.147)	(0.423)
Observations	293,838	122,495	171,343
Clusters (days)	184	184	184
R^2	0.0132	0.0056	0.0113
Adj. R^2	0.0131	0.0055	0.0113

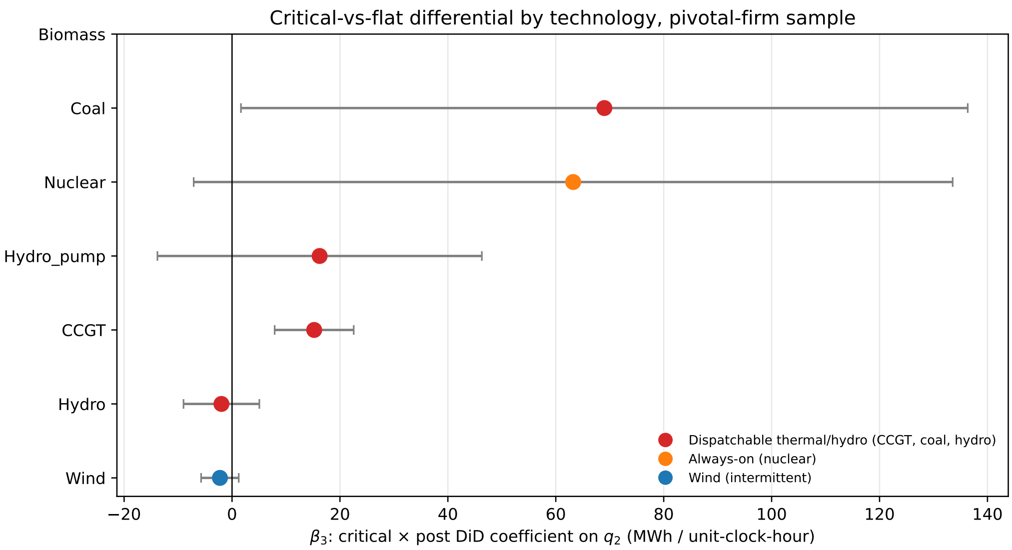


Figure 4: B1 – Heterogeneity in β_3 by technology, pivotal firms. Estimates of Equation 2 restricted to each technology in the pivotal-firm sample. Error bars are 95% cluster-robust confidence intervals. The dispatchable–non-dispatchable split is the key validation of the within-day-DiD design: granularity has economic content for technologies that can strategically reshape supply, not for must-run or price-taker technologies.

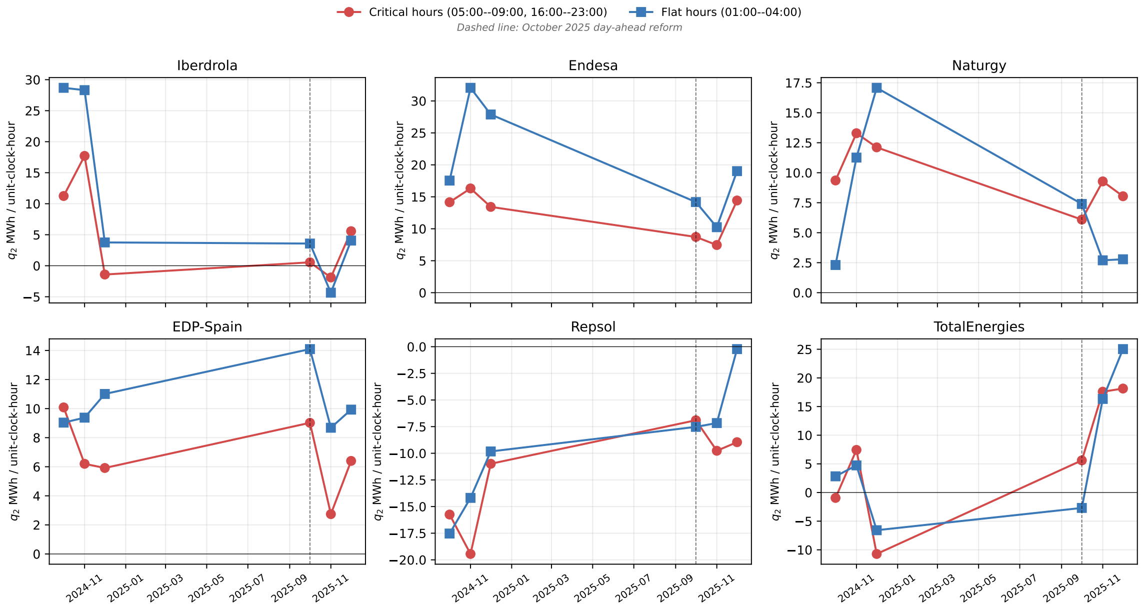


Figure 5: q_2 monthly mean by firm and hour-class, same-calendar-month windows. Six panels for the four dominant operators (Iberdrola, Endesa, Naturgy, EDP-Spain) and two non-pivotal placebo operators (Repsol, TotalEnergies). Red lines: critical hours (05:00–09:00 and 16:00–23:00). Blue lines: flat hours (01:00–04:00). Dashed vertical line: the October 2025 day-ahead reform.

The full table of β_3 stratified by technology (with the non-pivotal placebo column) is reported in Appendix A.4. The per-firm decomposition of β_3 jointly on q_2 and on day-ahead cleared volume (the withholding outcome) is in Appendix A.5. A supporting analysis on the day-ahead $\rightarrow$ intraday price wedge is in Appendix A.6; the wedge is a noisy outcome under the post-blackout *operación reforzada* regulatory overlay and is not informative on its own.

5.2 Parallel trends discussion

$$\mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid \text{crit}] \;=\; \mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid \text{flat}]\,. \tag{3}$$

6 Conclusion

A Robustness checks

A.1 Sensitivity to critical-hours definition

A.2 Sensitivity to firm-partition (pivotality vs administrative)

A.3 Same-cal-month vs full-window comparisons

Table 4: B5 – Robustness panel: β_3 across specifications, pivotal firms on q_2. Each row reports the headline coefficient β_3 under a single sensitivity. Section B5.1 varies the critical-hours definition between four candidates (price_peak is canonical). Section B5.2 varies the firm partition between the pivotality-based set (which includes EDP-PT) and the administrative IB/GE/GN/HC tier; the two coincide because EDP-PT lacks `pibci` data so its inclusion does not alter the q_2 regression. Section B5.3/B5.5 varies the window: the full 2024–2025 panel reduces the coefficient because the pre-MTU15-DA half of the panel is post-MTU15-IDA, where q_2 magnitudes are already crushed by the IDA reform; excluding the reforzada-active months (April–September 2025) recovers a coefficient close to the same-calendar-month estimate. Section B5.4 verifies that excluding individual units (EDP-PT, ABO2G) does not change the result.

Sensitivity	β_3	SE	p	G
B5.1 — Critical-hours definition				
supply_ramp h{7,8,16,17,18}	4.28***	(1.26)	0.0007	184
price_peak h{18-22}	5.68***	(1.49)	0.0001	184
demand_peak h{16-20}	5.13***	(1.36)	0.0002	184
joint h{7-8, 16-22}	4.77***	(1.31)	0.0003	184
B5.2 — Firm partition				
Pivotality-based firm set	3.87***	(1.15)	0.0008	184
Administrative IB/GE/GN/HC	3.87***	(1.15)	0.0008	184
B5.3 / B5.5 — Window				
Full window 2024-2025	3.42***	(0.80)	0.0000	730
Full window minus Apr-Sep 2025	4.00***	(0.85)	0.0000	575
B5.4 — Sample exclusions				
Drop EDP-PT	3.87***	(1.15)	0.0008	184
Drop ABO2G	3.87***	(1.15)	0.0008	184
B5.6 — DST transition days (CET ↔ CEST)				
Same-cal-month, drop DST transition days	4.00***	(1.15)	0.0005	182
Full window, drop DST transition days	3.47***	(0.80)	0.0000	726
B5.7 — DST regime separation (clock-hour semantics differ across CEST/CET)				
Same-cal-month, CEST days only (Oct 1–25)	−2.45	(1.93)	0.2028	50
Same-cal-month, CET days only (Oct 27–Dec 31)	6.30***	(1.34)	0.0000	132

A.4 β_3 stratified by technology

Full numeric form of Figure 4: β_3 stratified by `tech_group` for both the pivotal-firm sample and the non-pivotal negative control. The "Predicted" column shows the model's directional prediction: + for dispatchable strategic technologies (CCGT, Coal, Hydro, Hydro_pump), 0 for must-run (Nuclear) and price-taker (Wind, Biomass). Among pivotal firms, the dispatchable techs all show large positive β_3 and Wind shows the predicted null; non-pivotal firms should show $\beta_3 \approx 0$ across all technologies.

Table 5: B1 – β_3 stratified by technology, pivotal vs non-pivotal firms. Same specification as Table 3, restricted by technology and by the firm partition. Significance markers as in Table 3.

	CCGT	Hydro	Hydro pump	Coal	Nuclear	Wind	Solar PV
Critical × Post (β_3)	15.23	-1.963	16.24	69.00	63.22	-2.244	0
	(3.738)	(3.586)	(15.34)	(34.36)	(35.88)	(1.779)	(.)
Obs.	25,232	16,997	3,125	413	2,440	31,344	7,276
Clusters	184	184	183	58	118	184	184

A.5 Per-firm decomposition: q_2 and day-ahead cleared MWh

The mechanism storyboard predicts *negative* β_3 on day-ahead cleared volume (firms withhold in the day-ahead auction in critical hours post-reform) and *positive* β_3 on q_2 (firms add upward in the intraday). Per-firm the picture is mixed: Iberdrola and Naturgy show the joint negative-DA / positive-IDA pattern; Endesa shows positive-IDA without DA withholding; EDP-Spain shows weak effects on both. EDP-Portugal has no `pibci` entries (Portuguese-zone units do not appear in OMIE intraday programmes), so only the day-ahead side is identified for that firm.

Table 6: Per-firm β_3 on q_2 (B1) and DA cleared MWh (B3). Joint per-firm view of the strategic IDA upward adjustment outcome and the day-ahead withholding outcome.

	q_2 outcome (B1)		DA cleared (B3)	
Firm	β_3	SE	β_3	SE
Iberdrola	10.40^{**}	(3.25)	-62.37^{***}	(14.02)
Endesa	7.30^{*}	(3.06)	-2.22	(6.57)
Naturgy	2.58	(1.79)	-34.69^{***}	(7.69)
EDP-Spain	-2.29	(1.53)	-0.31	(3.33)
EDP-Portugal	—	—	-24.18^{*}	(11.43)

A.6 DA → IDA price wedge (supporting outcome)

The DA-to-IDA price wedge is a candidate supporting outcome but it is a noisy measure under the post-blackout *operación reforzada* regulatory overlay (April–September 2025). Same-calendar-month restriction (Oct–Dec 2024 vs Oct–Dec 2025) shows the within-day differential expanded modestly post-reform. The full-window specification attributes most of the within-day wedge spike to reforzada: once the reforzada × critical interaction is included, the post-MTU15-DA × critical coefficient becomes large and negative, indicating that the reform itself *reduces* the within-day wedge after controlling for the regulatory overlay. The wedge is therefore not informative on its own without reforzada controls.

Table 7: B2 – Within-day DiD on the DA → IDA price wedge. Outcome is hourly Spain DA clearing price minus IDA clearing price (averaged across IDA sessions) in EUR/MWh.

Specification	β_3	SE	p	G
Same-cal-month (Oct-Dec 2024 vs 2025)	4.62^{**}	(1.43)	0.0012	184
Full window, no controls	0.79	(1.18)	0.5043	730
Full window, + reforzada × crit	-10.06^{***}	(1.39)	0.0000	730
Full window, + reforzada + MTU15-IDA × crit	-10.06^{***}	(1.39)	0.0000	730

A.7 Operational vs strategic decomposition (PIBCA vs PHF)

A.8 Bid-shape robustness in the competitive-zone restriction

A.9 Time placebos

A.10 Triple-difference: pivotality as moderator

$$
\begin{aligned}
q_{2,fudh} = \alpha + \beta_1\,\mathrm{crit}_h + \beta_2\,\mathrm{post}_d + \beta_3\,\mathrm{piv}_f \\
+ \beta_{12}\,\mathrm{crit}_h\,\mathrm{post}_d + \beta_{13}\,\mathrm{crit}_h\,\mathrm{piv}_f + \beta_{23}\,\mathrm{post}_d\,\mathrm{piv}_f \\
+ \beta_{123}\,\mathrm{crit}_h\,\mathrm{post}_d\,\mathrm{piv}_f + \gamma_f + \delta_{\mathrm{DOW}(d)} + \boldsymbol{\theta}'\,X_{dh} + \varepsilon_{fudh},
\end{aligned}
\tag{4}
$$

where X_{dh} is the day-level vector of Spain wind, solar, and demand (GW) and enters in specifications (2)–(4) of Table 8. Specification (3) additionally interacts X_{dh} with each of crit_h, post_d, and piv_f; specification (4) absorbs clock-hour fixed effects.

Table 8: Triple-difference on q_2: crit $\times$ post $\times$ pivotal. Sample pools pivotal and non-pivotal firms across critical and flat hours, October–December 2024 and October–December 2025 (N=293,838 unit-day-hour cells). β_{123} is the headline triple-interaction coefficient; the implied within-group β_3 for pivotal and non-pivotal firms are reconstructions of columns (2) and (3) of Table 3. Standard errors clustered by date.

	(1) DDD	(2) + X	(3) + X×{crit, post, piv}	(4) + clock-hour FE
β_{123} (crit×post×piv)	8.605	8.623	8.612	8.619
	(1.362)	(1.364)	(1.361)	(1.364)
Implied β_3, pivotal firms	3.862	3.874	4.006	3.875
	(1.146)	(1.147)	(1.154)	(1.148)
Implied β_3, non-pivotal firms	-4.743	-4.748	-4.605	-4.743
	(0.423)	(0.422)	(0.488)	(0.422)
Observations	293,838	293,838	293,838	293,838

Table 9: Time-placebo β_3 on q_2. Same specification as Equation 2 re-estimated over windows that do not cross the October 2025 day-ahead reform. P1 and P2 are within-regime placebos; P3 shifts both windows back one year and crosses the June 2024 intraday reform. Pivotal-firm sample defined as in Table 1; non-pivotal placebo sample is the same negative-control firms used in the headline.

	P1: within-2024			P2: within-2025 pre-reform			P3: shifted back 1 yea		
	All	Pivotal	Non-piv.	All	Pivotal	Non-piv.	All	Pivotal	Non-
Critical × Post (β_3)	0.942	-0.0305	1.910	-0.0637	1.200	-1.133	-0.916	-2.785	0.1
	(0.490)	(1.235)	(0.440)	(0.407)	(0.894)	(0.280)	(0.536)	(1.341)	(0.4
Obs.	253,079	98,092	154,987	327,300	146,896	180,404	250,320	99,636	150,
Clusters	184	184	184	182	182	182	184	184	18

B Data appendix

B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

B.2 Firms in the analysis sample (detailed)

Extends the main-text Table 1 with a channel-segregated breakdown of each firm's 2025 cleared volume and the OMIE generation-unit count. Capacity is the proxy $\sum_u \max_{t\in 2025} \text{max\_power\_mw}(u,t)$ taken from the day-ahead bid headers (`cab`, sell side), share-weighted across joint-owned plants; it is a tight lower bound on installed nameplate.

I THINK WE CAN TAKE THE CAPAPCITY DATA! BUT MAYBE IT IS ONLY FOR EXPORTS AND IMPORTS!.

The three channel columns break the headline market share by the route through which each firm dispatches its production: *DA auction share* is the firm's share of Iberian DA-cleared volume in PDBF offer-type 1; *DA bilateral share* is the firm's share of bilateral contract executions in PDBF offer-type 4; *IDA cumulative share* is the firm's share of the final intraday-auction programme (PIBCA accumulated, last session per unit-period). The three channels differ in nature: the DA auction is anonymous and price-discovering, bilateral contracts are pre-negotiated firm-to-firm transfers (typically internal: from a group's generation arm to its retail subsidiary), and the IDA accumulates incremental adjustments after the DA clearing. Comparing channels reveals the routing strategy: Iberdrola is roughly balanced across channels, Endesa is auction-heavy (37% of DA-cleared), Naturgy is bilateral-heavy (16% of bilateral, only 4% of auction).

The net-position column uses cleared volumes in the day-ahead spot market for April–December 2025 (post-MTU15-IDA, so the pre-reform rule-28.8 opportunity-cost bids do not contaminate buy-side data); a firm is *net seller* if its cleared sales exceed 60% of its cleared day-ahead turnover, *net buyer* if below 40%, *mixed* otherwise. The aggregation joins generation units and retail-arm units of the same group via a broad `owner_agent` match. The label is descriptive only — vertical integration is not part of the identification strategy.

Table 10: Firms in the analysis sample (detailed): channel-segregated 2025 shares. The DA-auction, DA-bilateral and IDA-cumulative columns are constructed independently and do not sum to a single headline (their denominators are channel-specific). The right-most net-position column is included for descriptive completeness; the within-day DiD identification does not use vertical-integration status. Sources: OMIE unit register, `cab_all.parquet`, `pdbc_all.parquet`, `pdbf_all.parquet`, `pibca_all.parquet`.

Parent firm	Gen. units	Capacity (GW)	DA auction share (%)	DA bilateral share (%)	IDA cumul. share (%)	Net position
Dominant operators in Spanish generation (CNMC tier)						
Iberdrola	70	31.03	20.01	27.18	20.37	Mixed
Endesa	38	12.46	36.88	7.11	12.64	Net buyer
Naturgy	39	12.60	3.86	16.18	7.67	Net buyer
EDP-Spain	21	4.73	0.77	5.67	2.60	Net buyer
EDP-Portugal	16	7.42	20.62	12.70	5.17	—
Other large operators (non-dominant)						
Repsol	13	0.61	1.79	1.26	0.62	Net buyer
Engie España	59	4.05	0.03	2.49	1.07	Net seller
TotalEnergies	15	1.19	1.19	0.38	0.59	Net buyer
Moeve	112	1.28	0.03	—	0.66	Net buyer

B.3 File-format specifications and parser conventions

B.4 Power-vs-energy unit discipline

B.5 Coverage and missingness

B.6 Hour-of-day classification metrics

Table 11: Hour-of-day classification metrics, Spain 2025. For each clock-hour (Madrid local time), mean load, mean solar production, mean net-load (load − wind − solar), within-hour standard deviation of net-load σ_{within}, and mean signed within-hour load change $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$. Classification: **critical** hours have high σ_{within}; *flat* hours have low σ_{within}; *dropped* hours are borderline (moderate variance, no clean assignment).

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	Δ demand (MW)	Class
00:00–01:00	23.2	0.2	7.1	16.0	405	−864	*dropped*
01:00–02:00	22.4	0.1	6.8	15.4	353	−481	*flat*
02:00–03:00	21.9	0.1	6.6	15.1	269	−193	*flat*
03:00–04:00	21.9	0.1	6.5	15.3	324	194	*flat*
04:00–05:00	22.9	0.1	6.4	16.4	649	1179	*dropped*
05:00–06:00	24.9	0.3	6.3	18.3	766	1795	*critical*
06:00–07:00	27.0	1.9	6.2	18.9	1009	1420	*critical*
07:00–08:00	28.3	5.9	5.8	16.7	1259	602	*critical*
08:00–09:00	28.7	10.5	5.3	12.9	1116	218	*critical*
09:00–10:00	28.7	13.6	5.0	10.1	783	−103	*dropped*
10:00–11:00	28.6	15.0	4.8	8.7	576	−78	*dropped*
11:00–12:00	28.6	15.6	4.9	8.1	471	116	*dropped*
12:00–13:00	28.7	15.6	5.1	8.0	448	−60	*dropped*
13:00–14:00	28.3	15.2	5.3	7.9	453	−290	*dropped*
14:00–15:00	28.0	14.4	5.5	8.1	545	−57	*dropped*
15:00–16:00	28.3	13.0	5.8	9.4	747	344	*dropped*
16:00–17:00	28.9	10.1	6.3	12.5	1241	634	*critical*
17:00–18:00	30.0	6.4	6.8	16.8	1362	902	*critical*
18:00–19:00	31.2	2.7	7.3	21.2	1185	847	*critical*
19:00–20:00	32.0	0.7	7.4	23.9	557	163	*critical*
20:00–21:00	31.1	0.3	7.5	23.2	614	−1487	*critical*
21:00–22:00	28.6	0.2	7.5	20.9	801	−1926	*critical*
22:00–23:00	26.4	0.2	7.4	18.8	694	−1441	*critical*
23:00–00:00	24.6	0.2	7.2	17.2	534	−1093	*dropped*

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